Reza Nadimi

AI / ML Engineer · Vision-Language Models · RL · Agentic AI · Embedded & Robotics · Full-Stack
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Summary

AI/ML engineer (M.S. Mechatronics, University of Tehran) building agentic and vision-language systems end to end: RL fine-tuning (SFT → GRPO), tool-using agents, agent memory and calibrated evaluation, delivered through production full-stack, DevOps and mobile work. Hands-on background in embedded systems (STM32, PLC) and robotics control.

Education

University of Tehran — Tehran, Iran
M.S. in Mechatronics Engineering ***(GPA 19.27/20, 4.0/4.0)*** — *2025 – Present*

University of Tabriz — Tabriz, Iran
B.S. in Mechanical Engineering ***(GPA 18.25/20, 3.91/4.0)*** — *2021 – 2025*

Technical Skills

AI / ML: PyTorch, Hugging Face (Hub, Transformers), TRL, QLoRA, LangChain, LangGraph, LLM agents & tool calling, harness engineering, agent memory, retrieval (BM25, dense, hybrid RRF), SFT, GRPO, reinforcement learning (DQN, Dueling Double DQN, SAC), reward design, distillation, vision-language & multimodal models, YOLOv8, Faster R-CNN, ONNX Runtime, Stable-Baselines3, Gymnasium, MuJoCo, NumPy, Jupyter, LLM-as-judge, evaluation (bootstrap CIs, ECE calibration, uncertainty analysis), Gemini & OpenRouter APIs, LM Studio
Languages: Python, TypeScript, JavaScript, SQL, C, Java, MATLAB, Bash
Backend & Data: Django, DRF, FastAPI, Celery, RabbitMQ, Redis, PostgreSQL (RLS, pgTAP), SQLite, Elasticsearch, SQLAlchemy, Alembic, Drizzle ORM, Better Auth, Supabase (Auth, Storage, Edge Functions), Deno, gunicorn, WhiteNoise, REST APIs, SSE, transactional outbox, multi-tenancy, RBAC, rate limiting, Telegram Bot API
Frontend & Mobile: React, Next.js (App Router), React Native, Expo, TanStack Query, Redux, Context API, React Router, Tailwind CSS, Vite, React Three Fiber, Canvas2D, Web Workers, PWA & service workers, i18n with RTL, Android (Java)
Linux & DevOps: Linux, Bash & shell scripting, SSH, Git, GitHub, Docker, Docker Compose, Kubernetes, Kustomize, Argo CD (GitOps), nginx, GitHub Actions, GitHub Pages, Vercel, Cloudflare R2, Prometheus, CI/CD
Testing: Playwright, Vitest, pytest, pgTAP, Deno test, k6, axe accessibility testing, Django test framework, golden-file parity tests
Networking: Wireshark, nmap, TLS, DNS-over-HTTPS, SOCKS5, WebSocket, gRPC, CIDR scanning, Xray-core (VLESS, Trojan, VMess, Shadowsocks), DPI-evasion techniques, Clash / Sing-box formats
Cybersecurity: secure-by-design practices, threat modeling, least privilege, RBAC & row-level security, secrets management, rate limiting, CSP, input validation, supply-chain security (Trivy, pip-audit, cosign, Dependabot, digest pinning)
Embedded & Control: STM32F103 (HAL, CubeMX), ARM Cortex-M, Keil MDK-ARM, Proteus, Arduino, ESP32, GPIO, EXTI interrupts, UART/USART, SPI, I2C, ADC, timers, PWM, DMA, Siemens S7 PLC, STEP 7, PLCSIM, sensor fusion
Robotics & Simulation: MATLAB, Simulink, Simscape Multibody, inverse kinematics, PID, MPC, state feedback, observers, RRT
CAD/CAE, Thermo & CNC: SolidWorks, CATIA, Abaqus, fluid simulation, EES, thermodynamics (exergy), CNC programming

Experience

AI Researcher — 2025 – Present
AIA CodeHive (University Research Lab)
- Conduct research on vision-language and multimodal LLMs, focusing on **RL-based fine-tuning** for performance and alignment.
- Authored **two preprint papers** on agentic AI and harness engineering.
- Preparing **two further papers**: AI in thermodynamics and AI in robotics.

Automation, AI & Web Developer (Part-Time) — 2021 – 2025
ArjKhazaer
- Engineered AI fault-detection tools, including a **voice/noise-to-model pipeline** that converts machine noise into diagnoses.
- Shipped one diagnostic experience across web, Android and Windows desktop.

Intern, CNC Programming & Technology Department — Summer 2024 (3 months)
Iran Tractor Manufacturing Company (ITMCO) — *Tabriz, Iran*
- Worked in the **CNC programming and technology department** of the largest engine-components manufacturer, gaining hands-on experience in **automated machining**.

AI & Software Projects

Dental-Agent | Agentic VLM for Panoramic Radiographs — github.com/rezaxr14/VLM-DENTAL
Ongoing research · Qwen3.5, PyTorch, QLoRA, LangGraph, YOLOv8, GRPO, Hugging Face
- Architected an **agentic vision-language model** that reasons over multiple turns, calling deterministic tools (zoom/crop, contrast enhancement, FDI tooth numbering) and a learned detector before issuing a structured, FDI-grounded diagnosis.
- Built the data engine: verified multi-turn expert traces generated with **LangGraph** tool execution, cross-verified by independent model families with directive-leak checks, and versioned on **Hugging Face Hub**.
- Trained a cross-validated **YOLOv8** tooth detector, evaluated on held-out data, to replace oracle grounding in the live agent loop.
- Implemented the two-stage **SFT → GRPO** stack with **QLoRA** and a multi-objective reward (FDI accuracy, diagnosis, format adherence, tool validity and efficiency, LLM judge), configurable across compute tiers from cloud notebooks to local workstations.
- Built an ablation and evaluation harness (tool use vs. direct reasoning; GRPO vs. SFT vs. zero-shot frontier models) with bootstrap confidence intervals, **ECE calibration and uncertainty analysis**, failure analysis and trajectory visualization; shipped a unified CLI, YAML configs and an offline **pytest** suite.

RECAP | Memory & Recovery Engine for Tool-Using Agents github.com/rezaxr14/Recap

Python 3.12, FastAPI, PostgreSQL, Elasticsearch, SQLAlchemy/Alembic, Docker, GitHub Actions

- Architected an engine that records agent runs as idempotent, redacted event streams, compiles typed repairs from failures, validates them by *executing* them in a sandbox, and promotes them to memory only through a single **policy gate** with no bypass path.
- Built outcome-aware retrieval (**Elasticsearch** BM25 + dense + hybrid **RRF**) with compatibility checks and logged rejection reasons, so the system **fails closed** and returns "no safe memory" instead of a stale or incompatible repair.
- Implemented **MemoryShift** drift detection (downgrade and revalidate on environment change), a transactional outbox with retry worker, atomic index-alias rebuilds, and a **PostgreSQL** source of truth with **Alembic** migrations.
- Benchmarked against baselines with a safety audit; shipped a CLI, **FastAPI** HTTP API, end-to-end demo, threat model, ADRs and CI with integration tests on real PostgreSQL and Elasticsearch.

DentalForge | Multi-Tenant Observability & Annotation Platform github.com/rezaxr14/DentalForge

Next.js 16, React 19, TypeScript, PostgreSQL, Drizzle ORM, Better Auth, Redis, SSE, Cloudflare R2, ONNX Runtime Web, Canvas2D, Web Workers, Playwright, Vitest, k6, GitHub Actions, Vercel

- Architected and deployed a **multi-tenant** platform on **Next.js** and **PostgreSQL** for trace inspection, model evaluation and radiograph annotation, with admin/annotator/reviewer RBAC, tenant-scoped data access and Web Vitals budgets.
- Engineered a distributed worker architecture (versioned APIs, lease-based jobs, idempotent results, **Redis**-backed **SSE** with polling fallback) so **PyTorch** inference workers connect from local GPUs or cloud notebooks, with graceful degradation: remote worker → in-browser **ONNX** YOLO → stored replays → offline mode.
- Built LabelForge, an FDI-numbered annotation suite (**Canvas2D** editor, event-sourced undo/redo, optimistic autosave, conflict resolution, Cohen's κ reviewer agreement, active-learning queues, model-assisted pre-labeling, **YOLO/COCO** export).
- Developed TraceLab (virtualized trace exploration, turn-by-turn replay, perturbation auditing, tool vs. tool-free comparison) and an evaluation hub with bootstrap CIs, reliability diagrams, run diffs and reproducible leaderboards; ported metrics, GRPO reward components and all image tools from Python to **TypeScript** under golden-file parity tests.
- Established CI/CD on **GitHub Actions**: ADRs, typechecking, linting, unit and **Playwright** end-to-end tests, axe accessibility checks, production builds and worker-failure integration tests.

WhatToCook Smart Kitchen OS | Full-Stack AI Platform + GitOps github.com/rezaxr14/WhatToCook_Smart_Kitchen_OS
Demo: rezaxr14.github.io/WhatToCook_mobile_serverless

Django 5.2/DRF, React 18, TypeScript, TanStack Query, Celery, RabbitMQ, Redis, PostgreSQL (SQLite fallback), nginx, Docker, Kubernetes, Argo CD, GitHub Actions, Trivy, cosign, Playwright, LM Studio

- Architected and containerized a multi-service platform: **React** + **TypeScript** SPA (Vite, Tailwind, 3D landing, PWA) behind **nginx**, a **Django** REST API (gunicorn, WhiteNoise), **Celery** workers on a **RabbitMQ** broker, Redis and **PostgreSQL**, with migrations, seeding and static builds automated on boot via **Docker Compose**.
- Built a hybrid AI layer: camera-based multimodal pantry scanning, **SSE**-streamed recipe generation, an AI substitution assistant and a zero-waste matching engine (full and partial matches); moved generation to Celery and migrated it from a local LLM (**LM Studio**) to a hosted multimodal API while keeping a local-first fallback path.
- Engineered a dynamic JavaScript frontend that renders AI results without page reloads, a **Telegram bot** sharing pantry state across clients, and a 5-language UI with Persian/Arabic RTL; productionized with per-scope rate limiting (429 + Retry-After), a Django API test suite, an offline PWA and **Playwright** end-to-end tests.
- Shipped a GitOps release pipeline on **Kubernetes**: **Kustomize** overlays deployed by **Argo CD**, backend and frontend images pinned by digest with **cosign** verification, automated release-candidate promotion PRs and a **Prometheus** ServiceMonitor.
- Added a supply-chain security gate: **pip-audit**, **Trivy** image scans, scheduled security runs, digest-pinned base images and **Dependabot** across pip, npm, Docker and Actions; cleared fixable CVEs surfaced by the scans via a Pillow upgrade.

WhatToCook Serverless | Android + Web on Supabase github.com/rezaxr14/WhatToCook_mobile_serverless
Demo: rezaxr14.github.io/WhatToCook_mobile_serverless

React Native (Expo), React PWA, TypeScript, Supabase (Postgres, RLS, Edge Functions), Deno, pnpm, GitHub Actions

- Independently re-architected the product as a serverless stack with no servers to operate: an **Expo/React Native** Android app and an installable React **PWA** on one **Supabase** project, in a pnpm monorepo with shared types and AI contracts.
- Engineered the AI gateway as **Deno Edge Functions** (suggestions, streaming chat, vision pantry scans, substitutions) with a quota/cache path and pluggable Gemini, OpenRouter and LM Studio providers; provider keys never reach the client.
- Secured the data layer with Postgres **row-level security**, quota RPCs, validated `SECURITY DEFINER` RPCs and a SQL pantry-matching function, verified with **pgTAP** database tests, Deno function tests and mobile and shared unit tests.
- Automated delivery on **GitHub Actions**: typecheck, lint and tests on every change, staging deploys to Supabase, and an APK build per mobile change (verified on a physical Android device); guest sessions, camera scanning and five locales.

Xray Forge | DPI-Resistant Proxy-Config Toolkit + PWA github.com/rezaxr14/Xray_Forge
Demo: rezaxr14.github.io/Xray_Forge

Python, Xray-core, TLS, DNS-over-HTTPS, SOCKS5, Clash / Sing-box, JavaScript PWA, GitHub Actions, GitHub Pages

- Built a modular, pip-installable **Python** toolkit (GUI + CLI) for fixing, testing and exporting VLESS/Trojan/VMess/Shadowsocks configs: URI repair, templated naming, endpoint override, deduplication, subscription import and persistent settings.
- Implemented **DPI-resistant probing** (fragmented TLS ClientHello, decoy hello, record-layer fragmentation, domain fronting, SNI-case mixing) with TCP, TLS and WebSocket-upgrade checks, tunnel delay measurement and exit-country detection.
- Integrated real **Xray-core** through an auto-detected local **SOCKS5** bridge and built a CIDR scanner with anti-hijack DNS (**DNS-over-HTTPS** plus local resolvers) and GeoIP lookup; exports Clash Meta, Sing-box and base64 subscriptions.
- Engineering a cross-platform **DNS manipulation** layer with anti-censorship, anti-ads and gaming DNS profiles.

Embedded Systems, Robotics & Control Projects

Embedded Control | STM32 & Siemens PLC (STEP 7) github.com/rezaxr14/STM32_PLC_MECH2_embedded

STM32F103, C, HAL/CubeMX, Keil MDK-ARM, Proteus, STEP 7, PLCSIM

- Engineered **STM32F103** firmware (**HAL**, **CubeMX**, **Keil MDK-ARM**) across a broad peripheral set (GPIO, external interrupts, counters, timers, PWM, SPI, I2C, UART/USART, ADC), each design verified in **Proteus** simulation.
- Implemented peripheral drivers in all three I/O modes, **polling, interrupt and DMA**: UART/USART communication, ADC acquisition with averaging and voltage scaling, and timer-driven PWM with complementary channels, selectable polarity and output-compare timing; interfaced a 16x2 LCD in 4-bit mode and built multi-device communication and control.
- Programmed **Siemens S7 PLCs** in **STEP 7** and verified each automation problem in **PLCSIM**; documented every lab with schematics, simulation recordings and full engineering reports.

Autonomous Firefighting Robot github.com/rezaxr14/Autonomous-Firefighting-Robot

Arduino, IR sensors, servo motors, DC motors

- Built an autonomous robot on the **Arduino** platform that navigates, detects and extinguishes small-scale fires, using a three-sensor **infrared** flame array to locate the source.
- Integrated a **servo**-controlled water pump to aim and dispense water onto the detected flame, and programmed a dual-control system supporting fully autonomous operation and optional manual control.

Robotic Arm Control | Analytical IK vs. Reinforcement Learning github.com/rezaxr14/3D-Robot-Control-RL-Analytical

Python, PyTorch, NumPy, Gymnasium, Stable-Baselines3, MuJoCo

- Built a 3D simulation and control suite for a 3-link robotic arm with seeded reaching, moving-target and collision scenarios, optional **MuJoCo** physics and automated CI tests.
- Implemented a closed-form analytical **inverse kinematics (IK)** solver and trained a from-scratch **Dueling Double DQN** agent in **PyTorch** to reach 3D targets autonomously (0.70 held-out success, 7/10).
- Compared analytical IK, classical controllers (PD, DLS, MPC, RRT) and RL policies (DQN, SAC) on identical held-out seeds, identifying a 1.00 → 0.70 train-to-held-out gap.

Cart-Pole Control | Modern Control Design & Validation github.com/rezaxr14/cart-pole-modern-control

MATLAB, Simulink, Simscape Multibody

- Derived the **Lagrangian dynamics** and verified controllability and observability (Kalman rank, PBH); compared **PID**, **pole-placement** state feedback, integral tracking and full-/reduced-order **observers** on the nonlinear plant against settling-time, overshoot and steady-state specifications, including disturbance-rejection tests.

Solar Panel Cleaning Robot github.com/rezaxr14/solar-panel-cleaning-robot

Requirements-driven design and multibody simulation · SolidWorks, Simscape Multibody, MATLAB/Simulink, ESP32

- Derived actuation from requirements (25° slope, 10 mm obstacles, 50 mm panel misalignment, 1 mm accuracy): sized a 25.3:1 gearbox and a 12 N·m servo from static and dynamic analysis with explicit safety factors.
- Specified the **sensor-fusion** stack (0.062 mm encoder, ultrasonic step detection, IR edge detection, stereo depth camera) around an **ESP32** and validated the 28 kg **SolidWorks** model in **Simscape Multibody** trajectory simulations.

Thermodynamics & Mechanical Engineering Projects

CHP Plant with ORC Waste-Heat Recovery github.com/rezaxr14/gas-turbine-cogeneration-orc-waste-heat-recovery

Engineering Equation Solver (EES), thermodynamics

- Modeled a novel gas-turbine cogeneration (CHP) plant with an Organic Rankine Cycle (ORC) waste-heat recovery loop in EES.
- Performed first-law (energy) and second-law (**exergy**) analyses, quantified exergy destruction at component level to isolate the main sources of inefficiency, and validated the model against established academic references.

Solar–Biomass Polygeneration Plant github.com/rezaxr14/PowerPlant_Supercritical_CO2_KALINA

First-principles analysis · MATLAB (no toolboxes), EES

- Modeled and analyzed a coupled plant (oxy-gasifier, gas turbine with exhaust recirculation, molten-salt CSP, dual-pressure Rankine cycle, desalination, chlor-alkali, ammonia/urea synthesis) with hand-written fixed-point/Newton–Raphson solvers.
- Validated the model against the published reference design and cross-validated it against an independent EES model.

Finite Element Analysis of a Dental Implant github.com/rezaxr14/FEM-dental-implant

Abaqus, SolidWorks

- Modeled a multi-component dental implant assembly and simulated multiple occlusal (bite) load configurations; evaluated stress and strain distributions for different materials (e.g., titanium vs. zirconia) and validated results against academic literature.

Abaqus Structural Analysis github.com/rezaxr14/abaqus-structural-analysis

Abaqus, Python, MATLAB

- Four FEA studies (lap joint, composite shell buckling, plate deflection, membrane dome), each validated against closed-form analytical solutions with parametric sweeps and mesh-convergence checks.

Honors & Certifications

Ranked 1st (top student) in the 2025–26 master's cohort, University of Tehran College of Interdisciplinary Sciences & Technologies
Rank 35 (99th+ percentile), Iranian National Master's Entrance Examination in Mechanical Engineering
Esteddad-e Derakhshan (national program for exceptionally talented students); **National Elite Foundation** (Bonyad Nokhbegan) Vezvaii Reward
Harvard CS50 (edX): CS50x, AI, Python, Web, Cybersecurity
Faradars: Deep Learning, MATLAB, JavaScript, ReactJS, Java